

$$2) L(y, f(x)) = (1 - yf(x))_+ \\ = \begin{cases} 1 - yf(x) & , 1 > yf(x) \\ 0 & , 1 \leq yf(x) \end{cases}$$

$$E[L(y, f(x)) | X=x] = E[(1 - yf(x))_+ | X=x]$$

$$= p(x) E[(1 - f(x))_+ | X=x] + (1 - p(x)) E[(1 + f(x))_+ | X=x], \text{ where } p(x) = P(Y=1 | X=x) \\ = p(x)(1 - f(x))_+ + (1 - p(x))(1 + f(x))_+$$

we have:

$$(1 - f(x))_+ = \begin{cases} 1 - f(x) & , 1 > f(x) \\ 0 & , 1 \leq f(x) \end{cases}$$

$$(1 + f(x))_+ = \begin{cases} 1 + f(x) & , 1 > -f(x) = -1 < f(x) \\ 0 & , 1 \leq -f(x) = -1 \geq f(x) \end{cases}$$

	$1 - f(x)$	0
$1 > f(x)$	$1 > f(x)$	$1 \leq f(x)$
$1 + f(x)$	$-1 < f(x)$	$1 \leq f(x)$
0	$-1 \geq f(x)$	$f(x) \leq -1$
		No overlap

$\therefore 3$ outcomes: $f(x) \leq -1$; $-1 < f(x) < 1$; $f(x) \geq 1$

$$\Rightarrow E[L(y, f(x)) | X=x] = \begin{cases} p(x)(1 - f(x)) & , f(x) \leq -1 & ① \\ p(x)(1 - f(x)) + (1 - p(x))(1 + f(x)) & , -1 < f(x) < 1 & ② \\ (1 - p(x))(1 + f(x)) & , 1 \leq f(x) & ③ \end{cases}$$

Looking at each outcome:

$$① p(x)(1 - f(x)), f(x) \leq -1$$

$\therefore$ minimum when $f(x) = -1$ i.e. $f^*(x) = -1$

$$③ (1 - p(x))(1 + f(x))$$

$$= 1 + f(x) - p(x) - p(x)f(x)$$

$$= 1 + f(x) - p(x)(1 + f(x))$$

$\therefore$ Minimum when $f(x) = 1$ i.e. $f^*(x) = 1$

So, to minimize, f is bounded: $-1 \leq f(x) \leq 1$

$$② p(x)(1 - f(x)) + (1 - p(x))(1 + f(x))$$

$$= \cancel{p(x)} - p(x)f(x) + 1 + f(x) - \cancel{p(x)} - p(x)f(x)$$

$$= 1 + f(x) - 2p(x)f(x)$$

$$= 1 + f(x)(1 - 2p(x))$$

We see that if $p(x) > 0.5$, the expected loss decreases as $f(x)$ increases.

If $p(x) < 0.5$, the expected loss decreases as $f(x)$ decreases.

Hence, from all 3 cases, we have that:

If $p(x) > 0.5$, $f^*(x) = 1$

If $p(x) < 0.5$, $f^*(x) = -1$

$$\Rightarrow f_{\text{svm}}^* = \text{sign}[p(x) - \frac{1}{2}]$$